

- 13 The table shows assembly language instructions for a processor that has one register, the Accumulator (ACC).

| Label                                                                             | Instruction |           | Explanation                                                                                         |
|-----------------------------------------------------------------------------------|-------------|-----------|-----------------------------------------------------------------------------------------------------|
|                                                                                   | Opcode      | Operand   |                                                                                                     |
|                                                                                   | LDM         | #n        | Load the number n to the ACC                                                                        |
|                                                                                   | LDD         | <address> | Load the contents of the location at the given address to the ACC                                   |
|                                                                                   | LDI         | <address> | The address to be used is at the given address. Load the contents of this second address to the ACC |
|                                                                                   | ADD         | <address> | Add the contents of the given address to the ACC                                                    |
|                                                                                   | ADD         | #n        | Add the number n to the ACC                                                                         |
|                                                                                   | SUB         | <address> | Subtract the contents of the given address from the ACC                                             |
|                                                                                   | SUB         | #n        | Subtract the number n from the ACC                                                                  |
|                                                                                   | STO         | <address> | Store the contents of the ACC at the given address                                                  |
| <label>:                                                                          |             | <data>    | Gives a symbolic address <label> to the memory location with the contents <data>                    |
| # denotes a denary number, e.g. #123<br><label> can be used in place of <address> |             |           |                                                                                                     |

The current contents of memory are:

| Address | Contents |
|---------|----------|
| 563     | 125      |
| 899     | 63       |

